

2026-05-07-p2-f26-openhands-workspace

P2-F26 — Openhands Workspace + Task Planner + Step Executor

For agentic workers: Use `superpowers:subagent-driven-development`.

Programme position: F26 is the **sixth** Phase 2 feature (after F21-F25).

Goal: Ship container-based workspace management, a plan-tree-aware task planner, and a sequential step executor. Three new packages: `internal/workspace/`, `internal/planner/`. Reuses F25 plan trees and the `containers` submodule.

Spec: `docs/superpowers/specs/2026-05-07-p2-f26-openhands-workspace-design.md` **Q1-Q5:** Openhands / Core / Container-based / F25 plan tree + executor / Tools+slash+cobra

Zero new external deps — `digital.vasic.containers` already a submodule, `google/uuid` already direct.

Task list

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P2-F26-T01 — bootstrap F26 evidence + advance PROGRESS + CONTINUATION

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P2-F26-T02 — `internal/workspace/types.go` + `manager.go`: `Workspace`, `WorkspaceStatus`, `Manager` (TDD)

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P2-F26-T03 — `internal/workspace/workspace_tools.go`: `workspace_create/list/cleanup` (TDD)

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P2-F26-T04 — `internal/planner/types.go` + `executor.go`: `TaskStep`, `TaskPlan`, `StepExecutor` (TDD)

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P2-F26-T05 — `internal/planner/planner_tools.go`: `task_plan`, `task_step` (TDD)

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P2-F26-T06 — `/openhands slash + cobra` subcommands (TDD)

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P2-F26-T07 — `main.go` wiring + integration tests

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P2-F26-T08 — Challenge harness 5 phases + close-out + push 4 remotes

Task 1: Bootstrap F26

Append F26 header to 07_phase_2_evidence.md. Update PROGRESS.md: "F25 COMPLETE; F26 in flight". Update CONTINUATION.md: F26 mid-flight section.

Task 2: workspace/types.go + manager.go

internal/workspace/types.go: - Workspace struct (ID, Name, ContainerID, Image, ProjectDir, Status, CreatedAt, TTL) - WorkspaceStatus enum (Creating, Running, Stopped, Error) - Sentinels: ErrContainerRuntimeNotFound, ErrWorkspaceNotFound, ErrImagePullFailed - Constants: DefaultImage = "alpine:latest", DefaultTTL = 30 * time.Minute

internal/workspace/manager.go: - WorkspaceManager with ContainersOrchestrator interface for testability - Methods: CreateWorkspace(ctx, name, image, projectDir) (Workspace, error), ListWorkspaces() ([]Workspace, error), GetWorkspace(id string) (Workspace, error), CleanupWorkspace(ctx, id string) error - Container lifecycle: pull image → create container → start container - In-memory state tracking with sync.RWMutex - Container runtime detection: docker → podman → fallback error

Tests: real Docker/Podman gated (SKIP-OK if neither available); mock orchestrator for unit tests.

Task 3: workspace_tools.go

Three tools: workspace_create, workspace_list, workspace_cleanup. All implement tools.Tool. Category: workspace. RequiresApproval: create/cleanup → LevelEdit; list → LevelReadOnly.

Task 4: planner/types.go + executor.go

internal/planner/types.go: - TaskStep with StepType, Command, Prompt, Timeout, MaxRetries, Status, Output, Error - TaskPlan with ID, Name, WorkspaceID, Steps, Status - StepExecutor interface: ExecuteStep(ctx, step, workspace) error

internal/planner/executor.go: - SequentialExecutor implementing StepExecutor - ShellStep dispatch via exec.CommandContext in workspace container - LLMStep dispatch via llm.Provider.Generate - Retry logic: max retries with exponential backoff - Status tracking: Pending → Running → Completed/Failed

Tests with mock executor and real shell executor.

Task 5: planner_tools.go

Two tools: task_plan (creates a TaskPlan from an F25 PlanTree), task_step (executes a single step). Category: planner. RequiresApproval: both LevelEdit.

Task 6: /openhands slash + cobra

/openhands slash: status (list workspaces), create , cleanup helixcode workspace
{create,list,cleanup} cobra subcommands

Task 7: main.go wiring

Construct WorkspaceManager → Planner → Register tools → Register slash → Register cobra subcommands.

Task 8: Challenge harness

5 phases: - A: Create workspace → verify container running → cleanup → verify stopped - B: Create plan with F25 tree → execute shell step → verify output - C: Step retry on failure → verify retry count + eventual success - D: LLM step execution → verify response non-empty - E: Workspace list → verify count after create/cleanup

Plan written. Execute via TDD starting with T01.